

Lab Machine Results Comparison

January 12, 2026

1 Summary

This report evaluates the performance of four Large Language Models (LLMs) Phi-4, Mistral-7B-v0.3, Qwen2.5-7B, and Gemma-7B in generating a synthetic Q&A datasets.

Key Findings

- **Best Overall Performer:** Even though all of the models are almost on the same level Mistral-7B-Instruct-v0.3 is the recommended model. It demonstrated the best balance of speed, reliability, and competitive output quality.
- **Quality vs. Reliability Trade-off:** Qwen2.5-7B produced the highest semantic quality (BERT scores) but suffered from a critical failure rate, making it inefficient for automated pipelines without significant error handling.
- **Underperformance:** Gemma-7B consistently scored lower than its peers in both semantic accuracy and structural quality, despite having a moderate success rate.
- **Configuration Insight:** Increasing the `max_tokens` limit from 128 to 512+ statistically improved the output quality for most models. A minimum of 512 tokens is recommended to ensure structural integrity of the generated JSON data.

2 Methodology

- **Data Processing:** Text was parsed and split into chunks of varying sizes (128, 256, 512, 1024 tokens) using LangChain.
- **Sample Size:** 900 samples per chunk configuration.

- **Models Tested:**

- microsoft/Phi-4-mini-instruct
- mistralai/Mistral-7B-Instruct-v0.3
- Qwen/Qwen2.5-7B-Instruct
- google/gemma-7b

- **Evaluation Metrics:**

- *Efficiency*: Elapsed Time (minutes).
- *Reliability*: Success Rate (percentage of valid, parseable JSON outputs).
- *Quality*: BERT Score (F1) for semantic similarity and ROUGE-L for structural/phrasing overlap against reference standards.

3 Results and Analysis

3.1 Efficiency (Elapsed Time)

The computational cost generally increases with the `max_tokens` limit.

- Mistral-7B was consistently the fastest model, averaging around 200 minutes per run.
- Phi-4 and Qwen2.5 were slower, with Phi-4 averaging ≈ 270 minutes. This suggests Mistral is roughly 25–30% more efficient in terms of raw processing time for this specific task.

Table 1: Elapsed Time Comparison (Chunk Size 512, Max Tokens 2048)

Model	Elapsed Duration	Approx. Minutes
Mistral-7B-v0.3	4h 44m	284 min
Phi-4	5h 50m	350 min
Gemma-7B	7h 41m	461 min
Qwen2.5-7B	9h 47m	587 min

Data derived from run logs showing total generation time for the 2048 token limit configuration.

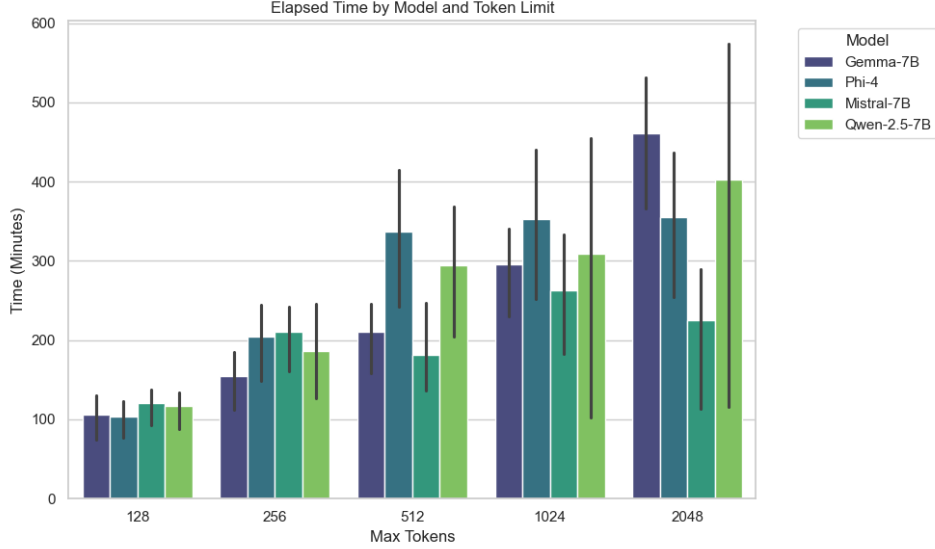


Figure 1: Elapsed Time Comparison

3.2 Reliability (Success Rate)

This metric tracks the model’s ability to strictly follow formatting instructions (e.g., generating valid JSON).

- **Winner:** Mistral-7B achieved the highest reliability. As shown in Table 2, Mistral encountered significantly fewer generation failures (23 failures) compared to Qwen (1,064 failures) at the same configuration.
- **Bottleneck:** Qwen2.5-7B struggled significantly, often failing to produce valid outputs.
- **Gemma-7B:** Maintained a moderate success rate but did not match the robustness of Mistral.
- **Phi-4:** Even though it’s struggling with small `max_tokens` it appears to be doing better than Gemma-7B at higher 512+ `max_tokens`.

Table 2: Raw Performance Metrics (Configuration: Chunk 512 / Max Tokens 1024)

Model	Success	Fail	BERT (F1)	ROUGE-L
Mistral-7B-v0.3	1599	23	0.586	0.193
Phi-4	1462	160	0.609	0.217
Gemma-7B	801	821	0.465	0.119
Qwen2.5-7B	243	1064	0.591	0.204

Note: "Fail" indicates the number of generations that could not be parsed into valid JSON. Qwen2.5 shows competitive BERT scores but a critically high failure count.

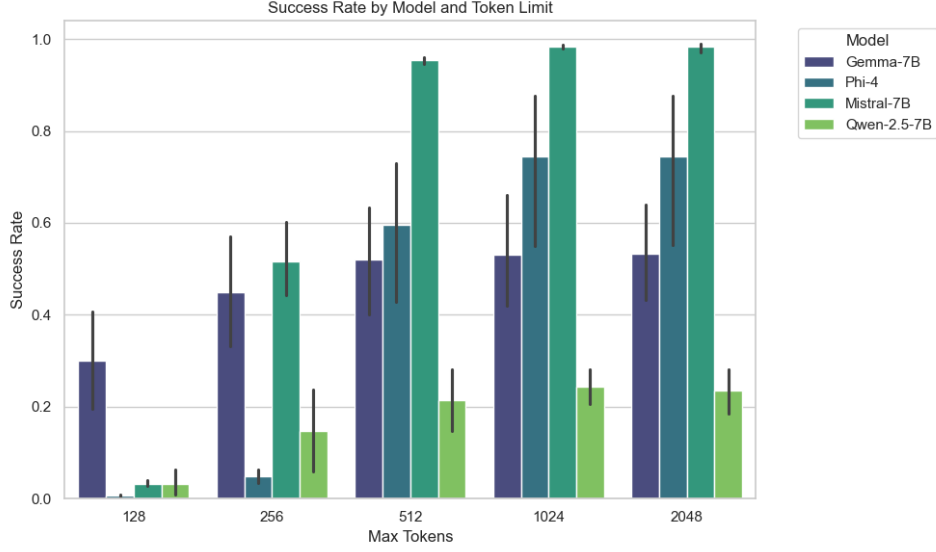


Figure 2: Success Rate Comparison

3.3 Semantic Quality (BERT F1 Score)

This metric measures how well the meaning of the generated Q&A matches the source context.

- **High Quality:** Qwen2.5-7B, despite its failures, produced high semantically accurate text, comparable to Phi-4.
- **Underperformer:** Gemma-7B consistently scored the lowest (approx. 0.45 – 0.50), indicating its generated questions and answers were less semantically aligned with the source text compared to the other three models.

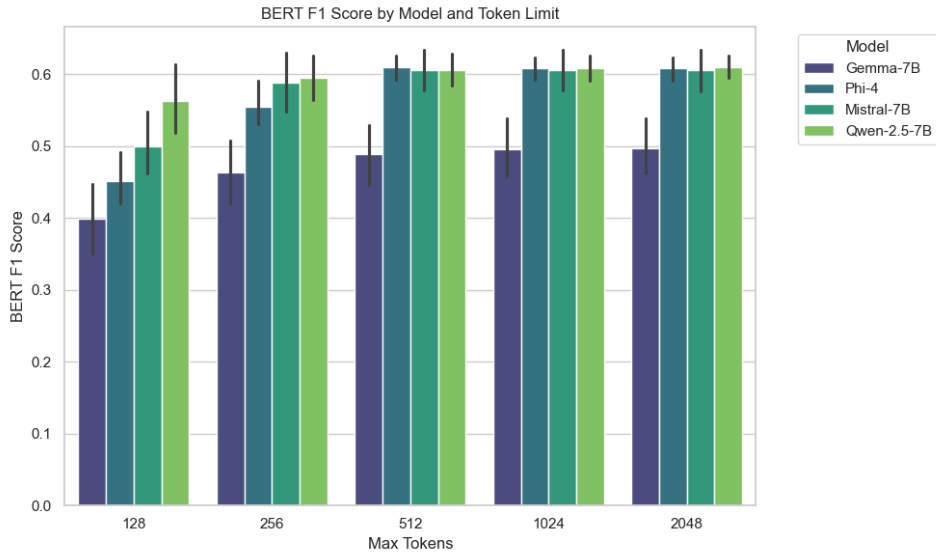


Figure 3: BERT F1 Semantic Quality Scores

3.4 Structural Quality (ROUGE-L Score)

This metric measures phrasing overlap and structural adherence.

- **Recovery of Phi-4:** While Phi-4 lagged slightly in semantic scores, it performed excellent structurally at higher token limits (512+), rivaling Mistral and Qwen.
- **Low-Resource Behavior:** Qwen2.5 was the only model to maintain decent structure at 128 tokens. Both Mistral and Phi-4 showed a sharp drop in quality at this low setting, confirming they require more “breathing room” (tokens) to form complete structures.
- **Gemma-7B Lag:** Gemma’s structural scores remained low across all token sizes, further confirming it struggled with the specific output format required.

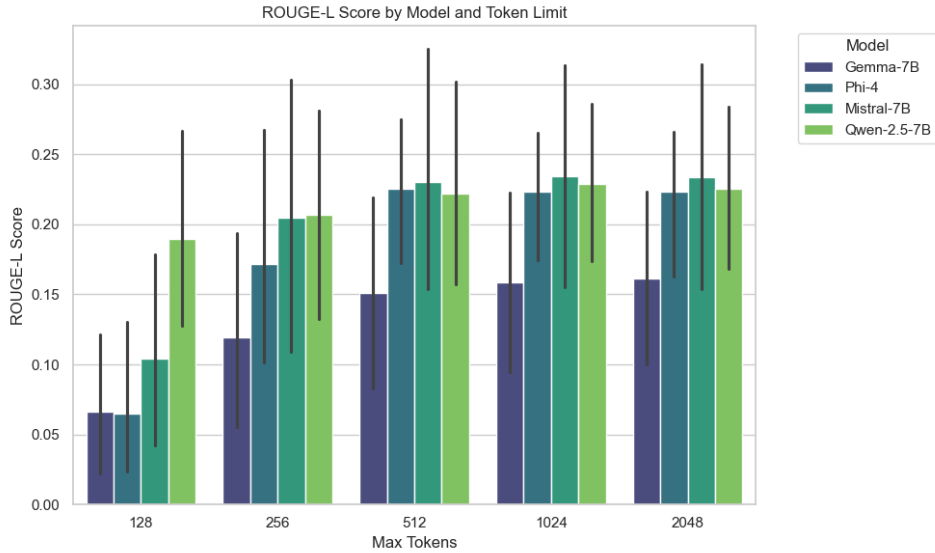


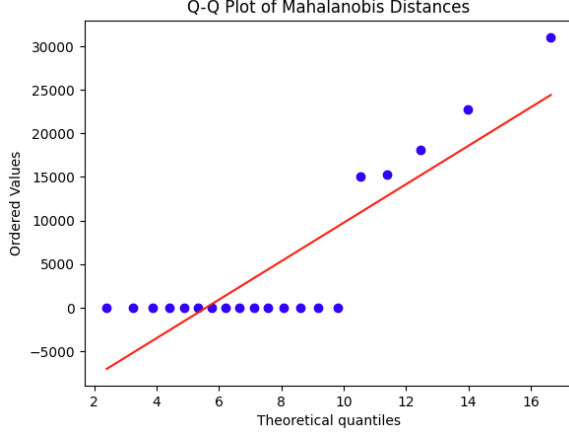
Figure 4: ROUGE-L Structural Quality Scores

4 Statistical Validation

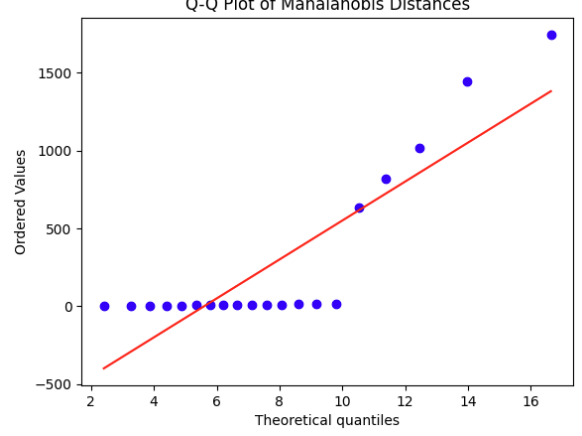
4.1 Multivariate Analysis Assessment (MANOVA)

To determine if the experimental configurations had a statistically significant combined effect on all quality metrics simultaneously, a Multivariate Analysis of Variance (MANOVA) was initially considered.

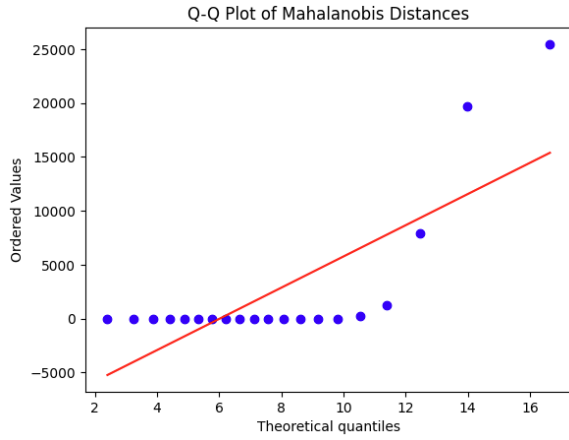
However, a critical assumption of MANOVA is *multivariate normality*—the requirement that the residuals of the dependent variables are normally distributed within each group. This assumption was tested using Quantile-Quantile (Q-Q) plots for each model.



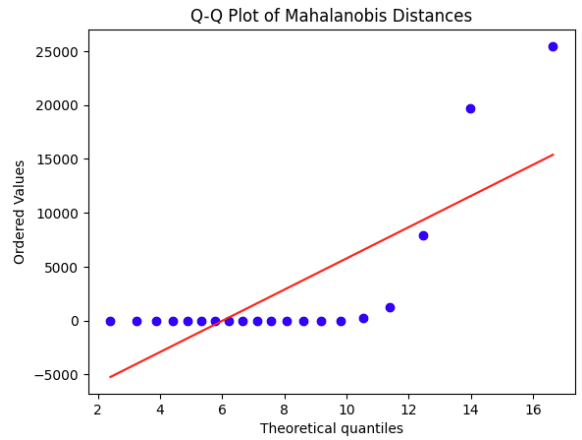
(a) Gemma-7B



(b) Phi-4



(c) Mistral-7B-v0.3



(d) Qwen2.5-7B

Figure 5: Q-Q Plots for Multivariate Normality Check. The significant deviation of data points from the theoretical normal distribution indicates a violation of the normality assumption required for MANOVA.

As illustrated in Figure 5, the data points deviate significantly from the red reference line, indicating a lack of normality in the distribution of the residuals.

Conclusion on MANOVA: Since the normality assumption is violated, the standard MANOVA test cannot be reliably used for this dataset. Instead, a non-parametric alternative such as **PERMANOVA** (Permutational Multivariate Analysis of Variance) is recommended for future multivariate testing, as it does not rely on the assumption of normality.

4.2 Pairwise Analysis (T-Tests)

Given the limitations of the multivariate approach on this specific dataset, we proceeded with robust Pairwise T-Tests with False Discovery Rate (FDR) correction to validate individual improvements.

Qwen2.5-7B (High Sensitivity):

Finding: Highly significant ($p < 0.05$) improvements in ROUGE scores when moving from 128 to 256+ tokens.

Implication: This model is severely penalized by tight token limits.

Mistral-7B & Gemma-7B (Moderate Sensitivity):

Finding: Statistically significant quality gains were observed when increasing tokens from 128 to 512.

Implication: 512 tokens is the statistical “minimum viable” setting for these models.

Phi-4 (Low Sensitivity):

Finding: The analysis showed no statistically significant difference in scores across token sizes.

Implication: Phi-4’s performance is relatively static; adding more tokens does not yield a statistically guaranteed improvement in quality, though the raw ROUGE plots suggest it helps with structure.

5 Conclusion and Recommendations

Based on the analysis of 7,000+ generated samples across four models:

Primary Recommendation: Deploy Mistral-7B-Instruct-v0.3

- **Configuration:** Chunk Size 512 or 1024 / Max Tokens 512 or 1024.
- **Reasoning:** It offers the highest probability of success (getting usable data) with the lowest computational time, while maintaining quality scores that are statistically competitive with the best model.

Secondary Option (High Fidelity): Qwen2.5-7B

- **Use Case:** If the project prioritizes semantic nuance over quantity/speed, and if the data pipeline can handle high failure rates (e.g., by implementing a retry mechanism).

Models to Avoid

- **Gemma-7B:** consistently underperformed in quality metrics.
- **Phi-4:** While capable, it is slower than Mistral and offers no quality advantage to justify the extra compute time.